Behavioral Attribution Module - (BAM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-AIG-BAS-BAM-2026-06-26-0003
Architecture Ecosystem: Structured Reality Standards™
Architecture Family: AI Governance Architecture (AIG®)
Operational Layer: AI Behavioral Governance Layer
Governed Space: Behavioral Attribution
Category: AI Governance
Subcategory: AI Behavioral Governance
Type: Behavioral Auditability Standard Module
Parent Standard: Behavioral Auditability Standard (BAS)
Version: 1.0
Status: Canonical · Open Module
Origin Date: 26 June 2026
Compatibility: OOF Methodology OS · AI Governance Architecture (AIG®) · Governance Architecture (GOA™) · Accountability Governance Architecture (AGA™) · Operational Reality Architecture (ORA™) · Cognitive Governance Intelligence Architecture (CLIA®) · Memory Governance Intelligence Architecture (MGIA™)
AI-Readable: Yes
Authority: OOF
Protection: MIP™ — Methodological Intellectual Property
Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Behavioral Attribution Module (BAM) defines the structural conditions under which AI behavior can be accurately attributed to its originating AI system, autonomous agent, human participant, governance authority, operational process, or external influence throughout the complete behavioral lifecycle.

BAM governs behavioral attribution.

The module establishes the foundational conditions required to ensure that every significant AI behavioral activity has a clearly identifiable origin, responsible actor, governance relationship, and attributable execution path.

Behavior may be reconstructed.

Behavior must also be attributable.

BAM governs that attribution.

Module Operational Role

BAM defines the behavioral-attribution layer of BAS.

Its role is to establish and preserve the identifiable origin and governance ownership of AI behavior.

Module Operational Space

BAM governs:

- behavioral attribution
- behavioral ownership
- behavioral origin identification
- actor attribution
- authority attribution
- Human-AI attribution
- multi-agent attribution
- governance attribution

The module applies wherever AI behavior must be linked to a specific origin, responsible entity, or governance authority.

Module Function

The module applies wherever systems must preserve:

- attributable AI behavior
- identifiable behavioral origin
- traceable governance ownership
- accountable behavioral execution
- governance-valid attribution
- trustworthy behavioral records

Its function is to ensure that every significant AI behavior can be reliably associated with the entity, authority, system, or process that produced or authorized it.

Minimum Implementation Framework

1. Define the Behavioral Attribution Object

The organization must define which AI behaviors require attribution.
This may include:

- autonomous decisions
- Human-AI interactions
- recommendations
- robotic actions
- multi-agent activities
- delegated decisions
- escalation activities

- safety-critical behaviors

2. Define Attribution Conditions

The system must define the information required to establish behavioral attribution.
This includes:

- originating AI system
- responsible authority
- delegated permissions
- operational context
- governance relationship
- execution identity

3. Define Attribution Failure Detection Logic

The system must identify attribution failures.
This may include:

- unknown behavioral origin
- ambiguous ownership
- missing execution identity
- broken attribution chain
- conflicting governance ownership
- governance-invalid attribution

4. Define Operational Response or Governance Logic

Governance response may include:

- attribution review
- ownership verification
- governance investigation
- authority reassessment
- accountability escalation
- behavioral correction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- attribution records
- governance ownership
- authority relationships
- behavioral reviews
- intervention history
- resulting behavioral states

An AI behavioral environment must not remain auditability-valid if materially significant behavioral attribution cannot be reconstructed, reviewed, validated, preserved, or governed.

Use Case 1 — Multi-Agent Financial Platform

Scenario

Several autonomous AI agents collaborate to evaluate investment opportunities before issuing a final recommendation.

Application

BAM attributes each behavioral contribution to the specific AI agent, delegated authority, governance role, and final decision owner.

Result

The organization strengthens accountability, improves governance transparency, reduces attribution ambiguity, and supports independent auditability.

Use Case 2 — Human-AI Manufacturing Environment

Scenario

A collaborative manufacturing system combines human decisions with autonomous robotic behavior during production.

Application

BAM attributes each operational behavior to either the human operator, autonomous robot, supervisory AI, or governance authority.

Result

The organization improves accountability, simplifies incident investigation, strengthens governance oversight, and increases trust in Human-AI collaboration.

Canonical Closing Statement

Behavioral Attribution Module (BAM) defines the structural conditions under which AI behavior can be accurately attributed to its originating AI system, autonomous agent, human participant, governance authority, operational process, or external influence throughout the complete behavioral lifecycle.

Behavior that can be reconstructed and verified must also have a clearly identifiable origin. Behavioral Attribution therefore becomes the third operational condition of Behavioral Auditability Standard within AI Governance Architecture (AIG®).

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

https://originopenfoundation.org/